

PH5102: INTRODUCTION TO BIOSTATISTICS IN ONE HEALTH

Effective Term

Semester A 2025/26

Part I Course Overview

Course Title

Introduction to Biostatistics in One Health

Subject Code

PH - Infectious Diseases and Public Health

Course Number

5102

Academic Unit

Infectious Diseases and Public Health (PH)

College/School

Jockey Club College of Veterinary Medicine and Life Sciences (VM)

Course Duration

One Semester

Credit Units

3

Level

P5, P6 - Postgraduate Degree

Medium of Instruction

English

Medium of Assessment

English

Prerequisites

Nil

Precursors

Nil

Equivalent Courses

Nil

Exclusive Courses

Nil

Part II Course Details

Abstract

Students will recognize the pivotal role of statistics in public health research and practice, and explore the application of statistical tools and methods to health and biological data collected through medical observations and epidemiological studies. They will acquire the basic knowledge and skills required for the collection, management, visualization, and analysis of health data, while learning key concepts in statistical inference and hypothesis testing. Students will demonstrate the ability to understand, describe, and visualize health data sets, design experimental studies, conduct basic statistical analyses, and attain the necessary foundation for learning more sophisticated statistical methods in the field of public health and epidemiology.

Course Intended Learning Outcomes (CILOs)

CILOs		Weighting (if DEC-A1 DEC-A2 DEC-A3 app.)			
1	Explain the key statistical concepts applied to public health		x	x	
2	Collect, organize, and interpret public health and epidemiological data		x	x	
3	Draw inferences about health-related parameters in different populations		x	x	x
4	Design various types of studies and analyse the resultant data		x	x	x

A1: Attitude

Develop an attitude of discovery/innovation/creativity, as demonstrated by students possessing a strong sense of curiosity, asking questions actively, challenging assumptions or engaging in inquiry together with teachers.

A2: Ability

Develop the ability/skill needed to discover/innovate/create, as demonstrated by students possessing critical thinking skills to assess ideas, acquiring research skills, synthesizing knowledge across disciplines or applying academic knowledge to real-life problems.

A3: Accomplishments

Demonstrate accomplishment of discovery/innovation/creativity through producing /constructing creative works/new artefacts, effective solutions to real-life problems or new processes.

Learning and Teaching Activities (LTAs)

LTAs		Brief Description	CILO No.	Hours/week (if applicable)
1	Lectures	Students will introduce to the fundamental concepts and methods applied in biostatistics, including data collection, organization, and visualization, descriptive statistics, sampling (sample size and power), key concepts of probability, estimation of population parameters (proportions and means), analysis of variance, correlation analysis, and basic statistical tests of hypothesis.	1, 2, 3, 4	2/h per week

2	Hands-on practical exercises	Students will engage in hands-on, problem-based exercises (data manipulation and analysis) to facilitate conceptual understanding, using JASP, R software and STATA	2, 3, 4	1/h per week
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Assessment Tasks / Activities (ATs)

ATs		CILO No.	Weighting (%)	Remarks ("- " for nil entry)	Allow Use of GenAI?
1	Classroom assessment	1, 2, 3, 4	20	Students will demonstrate their class attendance and active participation in the course.	No
2	Assignments	1, 2, 3, 4	40	Help to train students with basic knowledge, concepts, and analysis techniques.	Yes
3	Course Project	1, 2, 3, 4	40	Student will engage in a final course group project to apply the tools and techniques from the course.	Yes

Continuous Assessment (%)

100

Examination (%)

0

Minimum Continuous Assessment Passing Requirement (%)

40

Additional Information for ATs

To pass the course, students are required to obtain a minimum of 40% in continuous assessment and a minimum of 40% in the course project.

Assessment Rubrics (AR)**Assessment Task**

Classroom assessment (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

Criterion

Students will demonstrate their class attendance and active participation in the course.

Excellent

(A+, A, A-) High

Good

(B+, B, B-) Significant

Fair

(C+, C, C-) Moderate

Marginal

(D) Basic

Failure

(F) Not reaching basic levels

Assessment Task

Midterm examination (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

Criterion

Students will explain the concepts and topics taught in the classes (end of Week 6), communicate their knowledge in written format, and utilize relevant computer software to demonstrate their knowledge.

Excellent

(A+, A, A-) High

Good

(B+, B, B-) Significant

Fair

(C+, C, C-) Moderate

Marginal

(D) Basic

Failure

(F) Not reaching basic levels

Assessment Task

Final examination (for students admitted before Semester A 2022/23 and in Semester A 2024/25 & thereafter)

Criterion

Students will explain the concepts and topics taught in the classes from Week 7 to the end, communicate their understanding of the course content in written format, and utilize relevant computer software to demonstrate their knowledge.

Excellent

(A+, A, A-) High

Good

(B+, B, B-) Significant

Fair

(C+, C, C-) Moderate

Marginal

(D) Basic

Failure

(F) Not reaching basic levels

Assessment Task

Classroom assessment (for students admitted from Semester A 2022/23 to Summer Term 2024)

Criterion

The attendance and active participation of students in the classes and hands-on exercises.

Excellent

(A+, A, A-) Participation in >90% of the classes

Good

(B+, B) Participation in 85-90% of the classes

Marginal

(B-, C+, C) Participation in 70-85% of the classes

Failure

(F) Limited participation in classes (<70%)

Assessment Task

Midterm examination (for students admitted from Semester A 2022/23 to Summer Term 2024)

Criterion

The comprehension of the concepts and topics taught in the classes (end of Week 6), and ability to communicate that in the written format and using relevant computer software

Excellent

(A+, A, A-) Students achieve $\geq 86\%$ of the mark in the examination

Good

(B+, B) Students achieve ≥ 70 and < 86 of the mark in the examination

Marginal

(B-, C+, C) Students achieve ≥ 50 and < 70 of the mark in the examination (C letter grade is at least 50% or greater)

Failure

(F) Students achieve $< 50\%$ of the mark in the examination

Assessment Task

Final examination (for students admitted from Semester A 2022/23 to Summer Term 2024)

Criterion

The comprehension of the concepts and topics taught in the classes (from Week 7 to the end), and ability to communicate that in the written format and using relevant computer software

Excellent

(A+, A, A-) Students achieve $\geq 86\%$ of the mark in the examination

Good

(B+, B) Students achieve ≥ 70 and < 86 of the mark in the examination

Marginal

(B-, C+, C) Students achieve ≥ 50 and < 70 of the mark in the examination (C letter grade is at least 50% or greater)

Failure

(F) Students achieve $< 50\%$ of the mark in the examination

Additional Information for AR

To pass the course, students are required to obtain a minimum of 40% in continuous assessment and a minimum of 40% in the course project.

Part III Other Information**Keyword Syllabus**

Biostatistics; public health; data visualization, data analysis; study design; statistical inference

Reading List**Compulsory Readings**

Title	
1	Burt Gerstman. 2014. Basic Biostatistics: Statistics for Public Health Practice, Second Edition; Jones & Bartlett Learning. ISBN-13: 978-1284036015

Additional Readings

Title	
1	Wayne W. Daniel & Chad L. Cross. 2013. Biostatistics: A Foundation for Analysis in the Health Sciences, Tenth Edition, Wiley. ISBN-13: 978-1118302798 http://docshare02.docshare.tips/files/22448/224486444.pdf
2	Dohoo, Ian Robert, S. Wayne Martin, and Henrik Stryhn. 2012. Methods in Epidemiologic Research. Charlottetown, P.E.I.: VER, Inc.